Rule-State Inference (RSI): A Bayesian Framework for Compliance Monitoring in Rule-Governed Domains

Evidence from Francophone African Fiscal Systems

Abdou-Raouf Atarmla[1,2*]

[1]Institut National des Postes et Télécommunications (INPT), Rabat, Morocco

[2]Togo DataLab, Ministry of Digital Economy, Lomé, Togo

Emails: achilleatarmla@gmail.com | abdou-raouf.atarmla@datalab.gouv.tg

atarmla.abdouraouf@ine.inpt.ac.ma

2026

Abstract

Existing machine learning frameworks for compliance monitoring—Markov Logic Networks, Probabilistic Soft Logic, supervised models—share a fundamental paradigm: they treat observed data as ground truth and attempt to approximate rules from it. This assumption breaks down in *rule-governed domains* such as taxation or regulatory compliance, where authoritative rules are *known a priori* and the true challenge is to infer the *latent state* of rule activation, compliance, and parametric drift from partial and noisy observations.

We propose **Rule-State Inference (RSI)**, a Bayesian framework that inverts this paradigm by encoding regulatory rules as structured priors and casting compliance monitoring as posterior inference over a latent rule-state space $\mathbf{S} = \{(a_i,\, c_i,\, \delta_i)\}_{i=1}^{n}$, where a_i captures rule activation, c_i models the compliance rate, and δ_i quantifies parametric drift. We prove three theoretical guarantees: **(T1)** RSI absorbs regulatory changes in $O(1)$ time via a prior ratio correction, independently of dataset size; **(T2)** the posterior is Bernstein-von Mises consistent, converging to the true rule state as observations accumulate; **(T3)** mean-field variational inference monotonically maximizes the Evidence Lower BOund (ELBO).

We instantiate RSI on the Togolese fiscal system and introduce **RSI-Togo-Fiscal-Synthetic v1.0**, a benchmark of 2,000 synthetic enterprises grounded in real OTR regulatory rules (2022–2025). Without any labeled training data, RSI achieves F1 $= 0.519$ and AUC $= 0.599$, while absorbing regulatory changes in 0.2 ms versus 407 ms for full model retraining—a $2{,}000\times$ speedup.

Keywords: Bayesian inference, rule-governed domains, compliance monitoring, variational inference, zero-shot reasoning, fiscal compliance, Africa

*Independent work. Correspondence: achilleatarmla@gmail.com

1 Introduction

Machine learning has achieved remarkable progress in pattern recognition, forecasting, and classification. Yet in a wide class of high-stakes domains—tax administration, medical compliance, legal regulation, financial auditing—practitioners face a recurring structural difficulty: *the knowledge is already structured*. The rules are not unknown. They are written in law, codified in regulation, and enforced by institutions.

In these **rule-governed domains**, the challenge is not to *learn* the rules from data. The challenge is to infer *whether entities comply with them*, given that observations are partial, strategically manipulated, and frequently missing. A tax inspector does not need a neural network to determine that a company with 80 million FCFA in annual revenue is subject to VAT—the law says so. What she needs is a rigorous method to assess, from an incomplete set of declarations and behavioral signals, the actual compliance state of that company.

Limits of existing approaches. Supervised classifiers (XGBoost, neural networks) require labeled non-compliance examples, which are scarce, legally sensitive, and often unavailable in practice. Markov Logic Networks (MLNs) [Richardson and Domingos, 2006] and Probabilistic Soft Logic (PSL) [Bach et al., 2017] treat rule weights as quantities to be *learned* from data, inverting the epistemic direction. Neurosymbolic methods [Garcéz and Lamb, 2022] combine neural perception and symbolic reasoning, but their symbolic component remains a learned approximation. None of these frameworks provides: (i) zero-shot compliance assessment from authoritative rules, (ii) $O(1)$ adaptation to regulatory changes, nor (iii) native uncertainty quantification interpretable by non-technical auditors.

Our contributions. We propose **RSI (Rule-State Inference)**, a Bayesian framework that formalizes the inverse paradigm. Our contributions are:

(1) **The RSI framework**: a rigorous formalization of compliance monitoring as Bayesian posterior inference over a latent rule-state space, with rules as structured priors (Section 3).

(2) **Three theoretical guarantees**: $O(1)$ regulatory adaptability (T1), Bernstein-von Mises posterior consistency (T2), and monotone ELBO convergence (T3) (Section 4).

(3) **RSI-Togo-Fiscal-Synthetic v1.0**: a publicly released benchmark grounded in Togolese fiscal law, with a documented regulatory change event (Section 5).

(4) **Empirical validation** across five experiments demonstrating zero-shot performance, a $2{,}000\times$ update speedup, and robustness under 50% missing data (Section 5).

Why Francophone Africa. We ground our framework in the Togolese fiscal system for three reasons that generalize broadly. First, regulatory change frequency is high (multiple amendments per fiscal year), making adaptability not a luxury but a requirement. Second, data quality is structurally low: declared revenues are systematically under-reported (mean ratio 0.70 in our benchmark), and 18–20% of key fiscal variables are missing. Third, ML solutions designed for high-income contexts assume data richness that simply does not exist in these environments. RSI is designed *from* these constraints, not adapted *to* them.

2 Related Work

Probabilistic logic. MLNs [Richardson and Domingos, 2006] combine first-order logic with Markov random fields via weighted formulae, with weights learned from data. PSL [Bach et al., 2017] uses continuous truth values and achieves tractable MAP inference via convex optimization. Both treat rule weights as *outputs* of learning. RSI treats rules as *inputs*: their epistemic status is prior, not approximate. In MLNs, a zero-weight formula is effectively absent; in RSI, a rule with prior $\pi_i = 0.99$ is near-certain and overrides weak observational evidence.

Neurosymbolic AI. Garcez and Lamb [2022] describe architectures ranging from pipeline models to fully differentiable systems (Neural Theorem Provers [Rocktäschel and Riedel, 2017], LTNs [Badreddine et al., 2022]). The canonical direction is *learning for reasoning*: symbolic structure guides neural learning. RSI occupies the orthogonal position: *reasoning from rules*, where the symbolic layer is fixed and authoritative, not learned.

Interpretable rule learning. Bayesian Rule Lists [Letham et al., 2015] apply Bayesian model selection to rule sets—again, rules are the output. Our work produces not rules but *posterior distributions over compliance states*—a qualitatively different inference target.

AI and tax compliance. ML has been applied to tax gap estimation [Gomes et al., 2022], audit selection, and VAT fraud detection, universally under supervised paradigms. No prior work formalizes compliance monitoring as Bayesian state inference.

Missing data and low-resource learning. Frénay and Verleysen [2014] survey learning under label noise. RSI addresses a related but distinct challenge: *structural* data absence in low-integrity environments. Our likelihood handles missing values natively by assigning unit probability, preserving the prior without bias injection.

3 The RSI Framework

3.1 Problem Formulation

Let $\mathcal{R} = \{r_1, \ldots, r_n\}$ be a set of known regulatory rules. Each rule $r_i : \mathcal{X} \times \Theta_i \to \{0, 1\}$ maps entity attributes $x \in \mathcal{X}$ and rule parameters Θ_i to a binary applicability judgment. Let $\mathbf{D} = \{d_1, \ldots, d_m\}$ be a set of noisy observations of entity behavior.

Definition 1 (Rule-Governed Domain)**.** *A domain* $(\mathcal{R}, \mathcal{X}, \mathbf{D})$ *is* rule-governed *if: (i)* $\mathcal{R}$ *is authoritative and known a priori; (ii)* $\mathbf{D}$ *is partial, potentially missing, and may be strategically distorted; (iii) the primary inference task is to assess entity compliance with* $\mathcal{R}$*, not to learn it.*

3.2 The Latent Rule-State Space

Definition 2 (Rule State)**.** *The latent state of rule* r_i *is:*

$$s_i = (a_i,\ c_i,\ \delta_i), \quad \mathbf{S} = (s_1, \ldots, s_n) \in \mathcal{S}$$

where $a_i \in \{0, 1\}$ *indicates whether* r_i *is in force,* $c_i \in [0, 1]$ *is the compliance rate, and* $\delta_i \in \mathbb{R}$ *captures parametric drift.*

Core insight. This object has no equivalent in existing ML frameworks. MLNs have rule weights; RSI has rule states. The difference is epistemic: weights are inferred from data (data → rule); states are inferred given rules (rules + data → state).

3.3 The Prior $P(\mathbf{S})$: Rules as Structured Priors

Under mean-field factorization:

$$P(\mathbf{S}) = \prod_{i=1}^{n} P(a_i)\, P(c_i \mid a_i)\, P(\delta_i \mid a_i)$$

$$a_i \sim \text{Bernoulli}(\pi_i) \tag{1}$$

$$c_i \mid a_i{=}1 \sim \text{Beta}(\alpha_i, \beta_i) \tag{2}$$

$$\delta_i \mid a_i{=}1 \sim \mathcal{N}(0,\ \sigma_i^2) \tag{3}$$

Hyperparameters $(\pi_i, \alpha_i, \beta_i, \sigma_i)$ encode institutional knowledge: historical compliance rates, rule activation priors, and expected parametric stability.

3.4 The Likelihood $P(\mathbf{D} \mid \mathbf{S})$

Observations are conditionally independent given the rule state:

$$P(\mathbf{D} \mid \mathbf{S}) = \prod_{j=1}^{m} P(d_j \mid \mathbf{S})$$

For each observation d_j:

$$P(d_j \mid \mathbf{S}) = \sum_{i:\, r_i \text{ applicable}} \Big[a_i \cdot \mathcal{L}(d_j \mid r_i, c_i, \delta_i) + (1 - a_i) \cdot \varepsilon \Big]$$

Missing data handling: when d_j is absent, $P(d_j \mid \mathbf{S}) \equiv 1$, preserving the prior.

We use *segment-specific* likelihood functions by fiscal regime: (i) **TPU** (informal sector, turnover $<$ 30M FCFA): signals are payment delay, bank account formalization, and under-declaration ratio; (ii) **VAT** (turnover $\geq$ threshold): declared vs. theoretical VAT under a log-Gaussian noise model; (iii) **CIT** (turnover $\geq$ 100M FCFA): declared vs. theoretical corporate income tax given declared profits.

3.5 Posterior Inference

$$\boxed{P(\mathbf{S} \mid \mathbf{D}) \;\propto\; P(\mathbf{D} \mid \mathbf{S}) \cdot P(\mathbf{S})}$$

The posterior supports three interpretable queries:

- $P(a_i{=}1 \mid \mathbf{D})$: probability that r_i is active
- $\mathbb{E}[c_i \mid \mathbf{D}] \pm \sqrt{\mathrm{Var}[c_i \mid \mathbf{D}]}$: expected compliance with uncertainty
- $\mathbb{E}[\delta_i \mid \mathbf{D}]$: estimated parametric drift

All quantities are directly interpretable by non-technical auditors, without any post-hoc explanation tool.

3.6 Mean-Field Variational Inference

Exact inference is intractable for large n. We approximate the posterior by minimizing $\mathrm{KL}[Q_\phi(\mathbf{S}) \| P(\mathbf{S} \mid \mathbf{D})]$, equivalently maximizing:

$$\mathrm{ELBO}(\phi) = \mathbb{E}_{Q_\phi}[\log P(\mathbf{D} \mid \mathbf{S})] - \mathrm{KL}[Q_\phi(\mathbf{S}) \| P(\mathbf{S})]$$

All updates are *analytical* (conjugate families): Beta-Binomial for c_i, Bernoulli for a_i, Gaussian-Gaussian for δ_i.

Algorithm 1: RSI: Rule-State Inference (Mean-Field VI)

Input: Observations $\mathbf{D}$, Rules $\mathcal{R}$, Priors $\{\pi_i, \alpha_i, \beta_i, \sigma_i\}$
Output: Posterior $\{P(a_i|\mathbf{D}),\ \mathbb{E}[c_i|\mathbf{D}],\ \mathbb{E}[\delta_i|\mathbf{D}]\}$

```
Initialize Q_φ ← P(S);
repeat
    for each rule r_i ∈ R do
        Compute compliance signals from D;
        Update Q(a_i) ← Bernoulli(ρ_i);
        Update Q(c_i) ← Beta(α_i + n_ok, β_i + n_fail);
        Update Q(δ_i) ← N(μ_post, σ²_post);
    end
    Compute ELBO_t;
until |ELBO_t − ELBO_{t−1}| < ε;
return posterior summary
```

4 Theoretical Guarantees

4.1 T1: $O(1)$ Regulatory Adaptability

Definition 3 (Regulatory Update)**.** *A regulatory update $\mathcal{U}_k$ replaces r_k with r'_k, differing only in parameters $\Theta_k \to \Theta'_k$. The cost $\mathcal{C}(\mathcal{U}_k)$ counts the required operations.*

Theorem 1 ($O(1)$ Adaptability)**.** *For any regulatory update $\mathcal{U}_k$, RSI update cost is $\mathcal{C}^{\text{RSI}}(\mathcal{U}_k) = O(1)$, independently of $|\mathbf{D}|$ and n.*

Proof. The RSI prior factorizes as $P(\mathbf{S}) = \prod_i P(s_i|\Theta_i)$. After $\mathcal{U}_k$, the updated posterior satisfies:

$$P'(\mathbf{S} \mid \mathbf{D}) \;\propto\; P(\mathbf{S} \mid \mathbf{D}) \cdot \underbrace{\frac{P(s'_k \mid \Theta'_k)}{P(s_k \mid \Theta_k)}}_{\text{scalar, } O(1)}$$

This correction ratio depends only on the parameters of rule k. Evaluating two parametric distributions requires $O(1)$ operations.
Numerical stability. The ratio is applied to the *functional form* (the kernel) of the posterior, not to its absolute value. The normalization constant is updated analytically through conjugate families (Beta-Binomial, Gaussian-Gaussian), guaranteeing that the posterior remains a properly normalized distribution. There is no risk of variance explosion or numerical divergence. □

Corollary 1.1. *Over T regulatory updates: RSI: $O(T)$; supervised ML (full retrain): $O(T \cdot |\mathbf{D}| \cdot E)$; PSL/MLN: $O(T \cdot |\mathbf{D}| \cdot n)$. RSI dominates in high-frequency regulatory environments.*

4.2 T2: Bernstein-von Mises Consistency

Theorem 2 (BvM Consistency)**.** *Let $\mathbf{S}^*$ be the true rule state. Under conditions (C1) identifiability, (C2) $P(\mathbf{S}^*) > 0$, (C3) twice-differentiable log-likelihood, (C4) finite positive-definite Fisher information $\mathcal{I}(\mathbf{S}^*)$:*

$$\sqrt{m}\big(\hat{\mathbf{S}}_m - \mathbf{S}^*\big) \;\xrightarrow{d}\; \mathcal{N}\big(0,\, \mathcal{I}(\mathbf{S}^*)^{-1}\big) \quad (m \to \infty)$$

Proof sketch. (i) The normalized log-likelihood converges a.s. by the LLN. (ii) A second-order Taylor expansion around $\mathbf{S}^*$ yields a Gaussian approximation. (iii) Under (C2) and (C4), the prior is dominated by the likelihood as $m \to \infty$, giving the stated Gaussian limit [van der Vaart, 2000]. □

Corollary 2.1 (Prior Robustness)**.** *For any two RSI priors P_1, P_2: $\text{TV}(P_1(\mathbf{S}|\mathbf{D}_m),\, P_2(\mathbf{S}|\mathbf{D}_m)) \to 0$ as $m \to \infty$. Two experts with different initial beliefs converge to the same compliance assessment.*

4.3 T3: Monotone ELBO Convergence

Theorem 3 (Monotone ELBO)**.** *RSI coordinate ascent updates produce a monotonically non-decreasing ELBO sequence: $\text{ELBO}_{t+1} \geq \text{ELBO}_t$ for all $t \geq 0$.*

Proof. Each coordinate update minimizes $\text{KL}[Q_{\phi_i} \| P(s_i|\mathbf{D})]$ while holding other factors fixed. Since $\text{ELBO} = -\text{KL}[Q_\phi \| P(\mathbf{S}|\mathbf{D})] + \log P(\mathbf{D})$ and $\log P(\mathbf{D})$ is constant, each update cannot decrease the ELBO [Blei et al., 2017]. □

5 Experimental Evaluation

5.1 The RSI-Togo-Fiscal-Synthetic Dataset

We introduce **RSI-Togo-Fiscal-Synthetic v1.0**, a benchmark of 2,000 synthetic enterprises grounded in official OTR rules [OTR, 2024], structured in four layers:

Enterprise features Sector, region, declared turnover.

Latent ground truth (a_i, c_i, δ_i) per rule—the unobserved state RSI must infer.

Noisy observations Declared taxes, behavioral proxies (banking, e-invoicing, payment delays), with mean under-declaration ratio 0.70 and 18–20% missing data rates.

Binary labels Derived compliance labels for baseline evaluation.

The dataset encodes five fiscal rules and a documented **regulatory change event**: VAT threshold 60M → 100M FCFA (Law n°2024-007, 30 December 2024), enabling direct empirical validation of Theorem 1.
Dataset realism. The turnover distribution follows a log-normal law per segment (TPU, intermediate, large), whose aggregate approximates a *power law*—the universal signature of revenue distributions in real economies [Gabaix, 2009]. The 2,000 synthetic enterprises are not uniform: they reproduce the concentration structure typical of an African market (60% informal, 25% SMEs, 15% large taxpayers), ensuring that results are not artifacts of an oversimplified dataset.

5.2 Baselines and Experimental Setup

RBS Deterministic Rule-Based System. No uncertainty. Fragile to missing data.

XGBoost [Chen and Guestrin, 2016] Fully supervised (150 estimators, depth 4).

MLP Fully supervised (64-32-16, ReLU, early stopping).

Critical distinction. RSI operates in *zero-shot* mode: no labeled compliance examples are used. XGBoost and MLP are trained under full supervision.

5.3 Results

Table 1: Performance comparison. RSI uses *no labels*. †: full supervision required. **Bold**: best zero-shot performance.

Model	F1	AUC	Recall	Labels?	Update cost
RSI (ours)	**0.519**	**0.599**	**0.909**	**No**	0.2 ms
RBS	0.182	—	0.214	No	∼0 ms
XGBoost†	0.967	0.999	0.978	Yes	407 ms
MLP†	0.087	0.747	0.153	Yes	61 ms

EXP-1: Overall Performance. RSI achieves F1 = 0.519 and AUC = 0.599 without any labeled training data, substantially outperforming the Rule-Based System (F1 = 0.182). The high recall (0.909) is consistent with the requirements of fiscal surveillance systems, where missing a non-compliant entity (*false negative*) is generally more costly institutionally than a false alarm (*false positive*). RSI naturally adapts to this asymmetry through the calibration of the decision threshold τ, without modifying the inference algorithm. XGBoost achieves near-perfect F1 under full supervision but cannot operate in the zero-label setting.

Table 2: Update time and post-update F1 (2025 period).

Model	Update time	Post-update F1	Retrain needed?
RSI (ours)	**0.2 ms**	0.411	No
XGBoost	407 ms	0.954	Yes
MLP	61 ms	0.000	Yes

EXP-2: $O(1)$ **Adaptability (Theorem 1).** RSI updates in 0.2 ms—a $\mathbf{2{,}000}\times$ speedup over XGBoost. Over T annual updates (Togo: $T \approx 4$–6), cumulative savings grow linearly with T and exponentially with dataset size, confirming Corollary 1.

EXP-3: BvM Consistency (Theorem 2). RSI posterior uncertainty $\sigma[c_i|\mathbf{D}]$ decreases monotonically with sample size N, consistent with Theorem 2. For $N < 25$, RSI achieves competitive recall while XGBoost cannot operate (insufficient labeled examples).

EXP-4: Missing Data Robustness. Under 20% missing data (typical for Francophone Africa), RSI F1 degrades by only -0.003 ($< 1\%$), while the Rule-Based System loses $> 22\%$ F1. At 50% missing data, RSI remains functional (F1 = 0.521) while deterministic methods collapse.

EXP-5: ELBO Convergence (Theorem 3). The ELBO sequence is empirically monotone non-decreasing and converges within 7 iterations, confirming Theorem 3.

5.4 Interpretability: A Worked Example

For an enterprise with declared turnover 72M FCFA and no declared VAT, RSI outputs:

- $P(\text{VAT active} \mid D) = 0.91$ — rule very likely applies
- $\mathbb{E}[c_{\text{VAT}} \mid D] = 0.23 \pm 0.18$ — low compliance, high uncertainty
- $\mathbb{E}[\delta_{\text{VAT}} \mid D] = +2.4$M FCFA — positive threshold drift

An auditor acts directly on this output. *The posterior is the explanation*—no post-hoc tool (SHAP, LIME) is required.

6 Discussion

Generalizability. RSI is domain-agnostic. Immediate extensions include medical protocol compliance, anti-money-laundering, environmental regulation, and legal contract monitoring. Any domain satisfying Definition 1 is a candidate.

Limitations. Inference quality depends on the accuracy of prior hyperparameters. Misspecified priors bias results, though T2 guarantees asymptotic recovery. Segment-specific likelihood functions currently require domain expertise; future work will explore amortized inference for likelihood learning within the RSI structure.

Relationship to Kalman Filtering. RSI shares architectural parallels with the Kalman filter: the prior plays the role of the process model, the likelihood is the measurement model, and the posterior is the state estimate. This connection opens avenues for sequential RSI, tracking rule-state evolution over time.

Scaling. Mean-field VI scales linearly in n. For full tax codes (hundreds of rules), factor graph extensions via Belief Propagation handle rule dependencies while preserving scalability.

7 Conclusion

We have introduced **Rule-State Inference (RSI)**, a Bayesian framework that formally inverts the dominant paradigm in compliance monitoring. By treating authoritative regulatory rules as structured priors and casting compliance assessment as posterior inference over a latent rule-state space, RSI provides what no existing framework offers: zero-shot compliance monitoring, $O(1)$ regulatory adaptability, native uncertainty quantification, and full interpretability.

Three theorems establish RSI's foundations: $O(1)$ adaptability (T1), Bernstein-von Mises consistency (T2), and monotone ELBO convergence (T3). Experiments on RSI-Togo-Fiscal-Synthetic v1.0—grounded in real Togolese fiscal law—validate all three empirically and demonstrate a $2{,}000\times$ update speedup over retraining-based methods.

RSI is designed for the realities of rule-governed environments in the Global South: frequent regulatory changes, structural data scarcity, and institutional requirements for auditable, interpretable decisions. We release the full framework, dataset, and implementation to support reproducible research.

Future work: sequential RSI for longitudinal compliance tracking, multi-country extensions across the ECOWAS fiscal zone, and learned likelihoods via amortized variational inference.

Acknowledgments

The author expresses his sincere gratitude to **Togo DataLab** (Ministry of Digital Economy and Digital Transformation) and to the **Office Togolais des Recettes (OTR)** for their institutional support, access to domain problems, and invaluable expertise on the Togolese fiscal regulatory framework.

This work also benefited from the academic environment of the **Institut National des Postes et Télécommunications (INPT)** of Rabat. The author thanks his collaborators and mentors for the enriching discussions that helped consolidate the theoretical foundations of this framework.

Note: The opinions expressed in this document are those of the author and do not necessarily reflect the official positions of the institutions mentioned.

AI disclosure: The author used AI-assisted tools for manuscript drafting and LaTeX formatting. All scientific contributions, theoretical claims, experimental design, and conclusions are the sole responsibility of the author.

References

Bach, S., Broecheler, M., Huang, B., and Getoor, L. (2017). Hinge-loss Markov random fields and probabilistic soft logic. *Journal of Machine Learning Research*, 18(109):1–67.

Badreddine, S., d'Avila Garcez, A., Serafini, L., and Spranger, M. (2022). Logic tensor networks. *Artificial Intelligence*, 303:103649.

Blei, D. M., Kucukelbir, A., and McAuliffe, J. D. (2017). Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of KDD*, pp. 785–794.

Frénay, B. and Verleysen, M. (2014). Classification in the presence of label noise: a survey. *IEEE Transactions on Neural Networks and Learning Systems*, 25(5):845–869.

Gabaix, X. (2009). Power laws in economics and finance. *Annual Review of Economics*, 1(1):255–294.

Garcéz, A. d'Avila and Lamb, L. C. (2022). Neurosymbolic AI: The third wave. *Artificial Intelligence Review*.

Gomes, M., Batista, J., and Carvalho, J. (2022). Machine learning for tax gap estimation. *Expert Systems with Applications*, 198:116810.

Letham, B., Rudin, C., McCormick, T. H., and Madigan, D. (2015). Interpretable classifiers using rules and Bayesian analysis. *Annals of Applied Statistics*, 9(3):1350–1371.

Office Togolais des Recettes (2024). Loi n°2024-007 du 30 décembre 2024 portant loi de finances pour l'exercice 2025. Lomé, Togo.

Richardson, M. and Domingos, P. (2006). Markov logic networks. *Machine Learning*, 62(1–2):107–136.

Rocktäschel, T. and Riedel, S. (2017). End-to-end differentiable proving. In *NeurIPS*, pp. 3788–3800.

van der Vaart, A. W. (2000). *Asymptotic Statistics*. Cambridge University Press.